Johan Vester Dinesen | Curricum Vitae

Helsinki, Finland johan.dinesen@aalto.fi

Education

PhD., Mathematics — **Aalto University, Finland**
Advisors: Eimear Byrne, Camilla Hollanti, Ragnar Freij-Hollanti — Sep 2023 – Expected Jun 2027

MSc., Mathematics — **Aalborg University, Denmark**
Thesis: Non-adaptive group testing: residuation theory and disjunct matrices
Advisors: Matteo Bonini, Oliver W. Gnilke — Sep 2020 – Jun 2023

BSc., Mathematics — **Aalborg University, Denmark**
Thesis: Error correcting codes and upper bounds
Advisor: Oliver W. Gnilke — Sep 2017 – Jun 2020

Employment

Doctoral Researcher — Sep 2023 – Present
Aalto University, Finland

- Conducting research in algebraic coding theory and combinatorics, with a primary focus on classical coding theory, rank metric codes, and q-matroids.

Lecturer — Aug 2022
Aalborg University, Denmark

- Lectured a three week brush-up course in mathematics for incoming students at the Technical Faculty of IT and Design and Faculty of Engineering and Science.

Student Counsellor — Apr 2020 – Mar 2023
Department of Mathematical Sciences, Aalborg University, Denmark

- Counselling both incoming and current students in academic content, study environment and group work, study plans, dispensation, and job opportunities.

Teaching Assistant — Sep 2019 – Jun 2023
Aalborg University, Denmark

Papers

Almost Affine Vector Rank-Metric Codes
with Matteo Bonini
Preprint: arXiv:2605.28072, 2026, 26 pages. Submitted to Advances in Mathematics of Communications.

A q-Polymatroid Framework for Information Leakage in Secure Linear Network Coding
with Eimear Byrne, Ragnar Freij-Hollanti, Camilla Hollanti
Preprint: arXiv:2601.07567, 2026, 28 pages. Submitted to IEEE Transactions on Information Theory.

The Star Product of Uniformly Random Codes
with Ragnar Freij-Hollanti, Camilla Hollanti, Benjamin Jany, Alberto Ravagnani
Preprint: arXiv:2511.17236, 2025, 30 pages. Submitted to Designs, Codes and Cryptography.

Secret Sharing in the Rank Metric
with Eimear Byrne, Ragnar Freij-Hollanti, Camilla Hollanti
2025 IEEE International Symposium on Information Theory (ISIT)

Group testing via residuation and partial geometries
with Oliver W. Gnilke, Marcus Greferath, Cornelia Rößing
Advances in Mathematics of Communications, 2025, 19(2): 397-405

Fellowships and Awards

- **Doctoral Research Grant (22,000€)** Sep 2026 – Apr 2027
  Foundation for Aalto University Science and Technology
- **Marie Skłodowska-Curie Fellow** Sep 2023 – Aug 2026
  Aalto University
- **ISIT 2025 Student Travel Grant (600$)** Jun 2025

Conference/Seminar/Colloquium Organization

- **Parallel Session on Algebraic and Combinatorial Coding Theory** Jan 8–9, 2026
  Session organized at the Finnish Mathematical Days 2026, Jyväskylä, Finland.
- **Workshop on Secure and Private Distributed Computation** May 27–28, 2024
  A 2-day workshop on algebraic and information theoretic methods in secure and private distributed computation.

Invited Talks

- **The star product of uniformly random codes** May 11, 2026
  Lunch Seminar Series at Aalborg University, Department of Mathematical Sciences
- **q-Matroids and secret sharing** Jul 7, 2025
  SIAM Conference on Applied Algebraic Geometry
- **Secret sharing in the rank-metric** Apr 9, 2025
  Helsinki CS Theory Seminar
- **Rank-metric codes and secret sharing** Apr 8, 2025
  Postgraduate International Coding theory Seminar

Contributed Talks

- **The Star Product of Uniformly Random Codes** Jun 26, 2026
  European Network in Coding Theory and Applications Workshop 2026
- **Group testing via residuation and partial geometries** Mar 5, 2026
  CIMPA School, Algebraic Geometric Methods in Information Theory, Santa Fe, Argentina
- **The star product of uniformly random codes** Jan 8, 2026
  Finnish Mathematical Days 2026, Jyväskylä, Finland
- **Group testing via residuation and partial geometries** Sep 6, 2025
  Finite Geometries 2025 – Seventh Irsee Conference, Irsee, Germany
- **On q-polymatroids and linear network coding** Jul 16, 2025
  Summer School on Contemporary Coding Theory and Applications, Munich, Germany
- **Secret sharing in the rank-metric** Jun 26, 2025
  IEEE International Symposium on Information Theory, Ann Arbor, Michigan, USA
- **Secret sharing in the rank-metric** Feb 10, 2025
  ENCODE Training School 3, Neuchâtel, Switzerland
- **q-Access structures and rank-metric codes** May 29, 2024
  ENCODE Training School 2, Paris, France
- **Secure Distributed Matrix Multiplication** Dec 11, 2023
  ENCODE Training School 1, Dublin, Ireland
- **Non-adaptive group testing and residuation theory** Sep 21, 2023
  ANTA Seminar, Aalto University

Research Visits / Schools

- **Research Semester at Aalborg University** — Jan 2026 – Jun 2026
  with Matteo Bonini
- **CIMPA School: Algebraic Geometric Methods in Information Theory** — Mar 2026
  Universidad Nacional del Litoral, Santa Fe, Argentina
- **Secondment at University College Dublin** — Oct 2024 – Jan 2025
  with Eimear Byrne
- **Visit at University College Dublin** — Mar 2024
  with Eimear Byrne
- **IEEE European School of Information Theory** — Jul 2024
  Eindhoven University of Technology, Eindhoven, The Netherlands

Industrial Secondment

- **Xiphera, Finland** — May 2025 – Aug 2025
  Error detection and correction for hardware-constrained systems (lightweight SEC-DED), analysis of code-based post-quantum cryptography (HQC, NIST), and advanced tutorials on finite fields and modern ECC (RS, RM, LDPC).

Teaching Experience

At Universidad Nacional del Litoral, Argentina

Teaching Assistant

– *Finite Fields and Coding Theory* with Professor Daniel Panario. Intensive course at the CIMPA School *Algebraic Geometric Methods in Information Theory.*

At Aalto University, Finland

Primary Instructor

– Galois Theory, Applications of Coding Theory to Security, Differential and Integral Calculus 1, Differential and Integral Calculus 2, Foundations of Discrete Mathematics

At Aalborg University, Denmark

Primary Instructor

– Calculus, Linear Algebra, Linear Algebra and Probability Theory, Linear Algebra with Applications, Introduction to Mathematical Methods, The Theoretical Foundations of Computer Science, Integrated Design-Engineering I: Mathematics, Spatial Structures and Parametric Design, Integrated Design-Engineering II: Mathematics, Spatial Structures and Parametric Design, Scientific Computing, Ordinary Differential Equations, Math101, Brush-up.

Services and Outreach

- **Workshop Instructor, Castleknock Community College, Dublin, Ireland** — 2025
  Delivered an introductory workshop on coding theory for secondary school students.
- **Peer Reviewer** — 2024 – Present
  Designs, Codes and Cryptography, IEEE International Symposium on Information Theory (ISIT), Advances in Mathematics of Communications
- **Elected Student Representative on the Department Council, Department of Mathematical Sciences, Aalborg University** — Feb 2021 – Jan 2023
  The task of the department council is to discuss and advise the department management on issues related to the department's research, education, knowledge dissemination and possibly public sector services.
- **Elected Student Representative on the Study Board, Department of Mathematical Sciences, Aalborg University** — Feb 2020 – Jan 2023
  The study board is responsible for ensuring the quality, organisation, implementation and development of the degree programmes and teaching affiliated to the study

board as well as for ensuring active staff and student involvement.

- **Founder/President of Den Diskrete Forening at Aalborg University** May 2018 – May 2022
  Den Diskrete Forening is a local student-driven society at The Department of Mathematical Sciences at Aalborg University. It promotes the socialising between both students and personnel at the department.
- **Orientation Mentor, Aalborg University** Sep 2018 – May 2023
  Helping new students at The Department of Mathematical Sciences at Aalborg University with social connections via tours and activities, administrative assistance, emotional support, logistics, and day-to-day.

Skills

Computer languages: Magma, SageMath, Python

Languages: English (fluent) / Danish (Native)